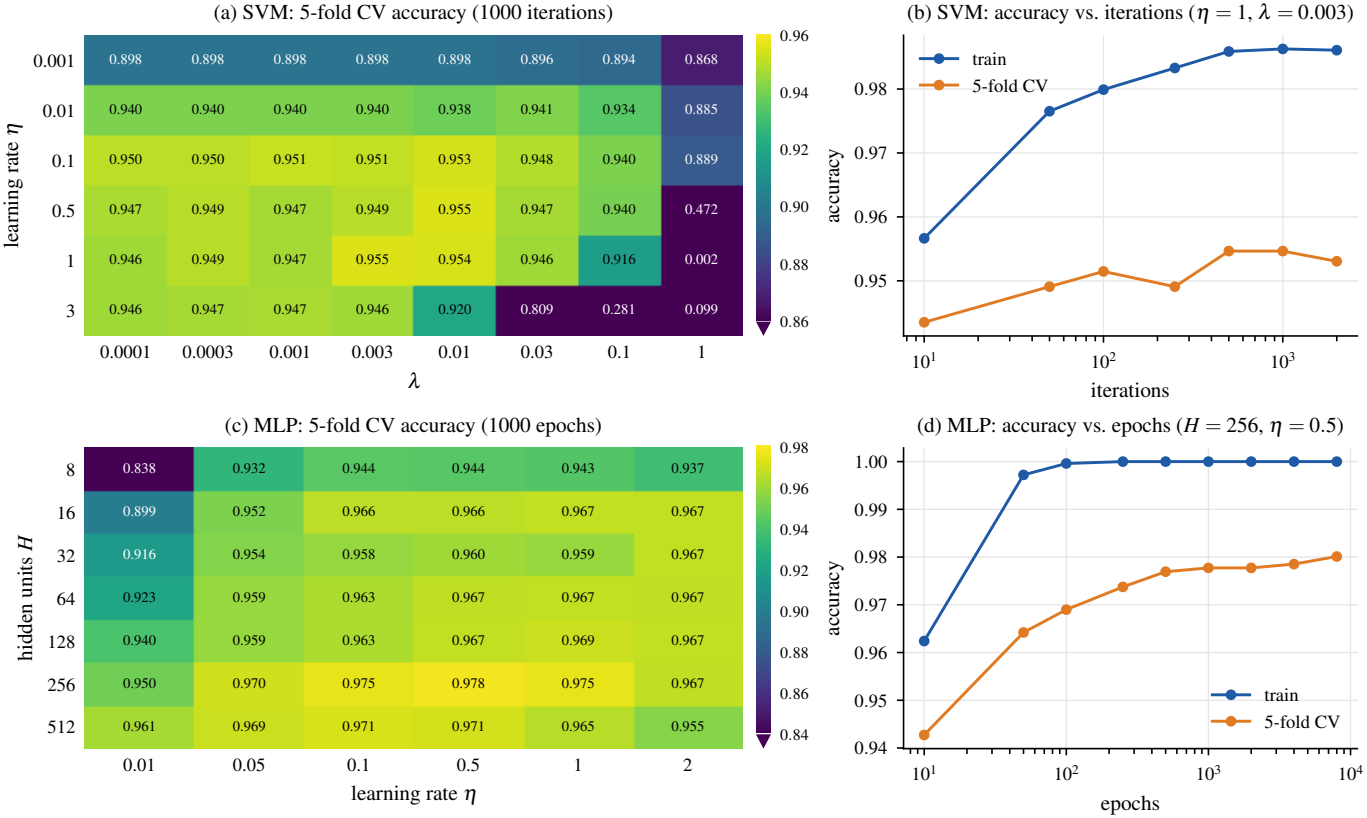
(a) SVM: 5-fold CV accuracy (1000 iterations)
learning rate η
0.001 0.898 0.898 0.898 0.898 0.898 0.896 0.894 0.868
0.01 0.940 0.940 0.940 0.940 0.938 0.941 0.934 0.885
0.1 0.950 0.950 0.951 0.951 0.953 0.948 0.940 0.889
0.5 0.947 0.949 0.947 0.949 0.955 0.947 0.940 0.472
1 0.946 0.949 0.947 0.955 0.954 0.946 0.916 0.002
3 0.946 0.947 0.947 0.946 0.920 0.809 0.281 0.099
0.0001 0.0003 0.001 0.003 0.01 0.03 0.1 1
λ
0.96 0.94 0.92 0.90 0.88 0.86
(b) SVM: accuracy vs. iterations ($\eta = 1$, $\lambda = 0.003$)
train
5-fold CV
accuracy
0.98 0.97 0.96 0.95
10^1 10^2 10^3
iterations
(c) MLP: 5-fold CV accuracy (1000 epochs)
hidden units H
8 0.838 0.932 0.944 0.944 0.943 0.937
16 0.899 0.952 0.966 0.966 0.967 0.967
32 0.916 0.954 0.958 0.960 0.959 0.967
64 0.923 0.959 0.963 0.967 0.967 0.967
128 0.940 0.959 0.963 0.967 0.969 0.967
256 0.950 0.970 0.975 0.978 0.975 0.967
512 0.961 0.969 0.971 0.971 0.965 0.955
0.01 0.05 0.1 0.5 1 2
learning rate η
0.98 0.96 0.94 0.92 0.90 0.88 0.86 0.84
(d) MLP: accuracy vs. epochs ($H = 256$, $\eta = 0.5$)
accuracy
1.00 0.99 0.98 0.97 0.96 0.95 0.94
train
5-fold CV
10^1 10^2 10^3 10^4
epochs